AI-Powered Intrusion Detection System (CyberShield IDS)

Full-Stack Security Dashboard · Personal Project

Project Summary

Designed and developed a real-time AI-powered Intrusion Detection System that monitors network traffic, classifies cyber threats using machine learning concepts, and presents actionable security insights through an interactive SOC-style dashboard. The system simulates a complete security pipeline — from data ingestion and feature extraction to threat classification, alert generation, and visual analytics.

What I Built

- A **real-time traffic simulation engine** that generates realistic network packets (TCP, UDP, ICMP, HTTP, SSH, DNS) with randomized source/destination IPs, ports, protocols, and payload sizes.
- An **ML-inspired classification layer** that categorizes traffic into Normal, DoS, Probe, R2L, and U2R attack types with confidence scores and severity levels modeled after the NSL-KDD dataset.
- A **live monitoring dashboard** with animated packet-level visualization, scanline effects, and real-time table updates using Framer Motion transitions.
- An **alert management system** with filtering by severity (low/medium/high/critical) and attack type, bulk acknowledge, and chronological sorting.
- An **analytics page** with interactive charts — attack distribution (PieChart), traffic flow over time (AreaChart), and protocol breakdown — built with Recharts.
- A **model insights page** displaying accuracy (96.5%), precision, recall, F1 score, a confusion matrix heatmap, and feature importance bar charts to simulate Random Forest classifier output.
- A fully **responsive, dark-themed UI** with a cybersecurity aesthetic, custom design tokens, and a persistent sidebar navigation.

Technical Skills Used

- ■ **Frontend:** React 18, TypeScript, Vite, Tailwind CSS, Framer Motion
- ■ **UI Components:** Radix UI, shadcn/ui, custom design system with HSL tokens
- ■ **Data Visualization:** Recharts (AreaChart, PieChart, BarChart, custom tooltips)
- ■ **State Management:** React hooks (useState, useEffect, useCallback), custom hooks
- ■ **Routing:** React Router DOM with multi-page SPA architecture
- ■ **ML Concepts:** Classification metrics, confusion matrix, feature importance, NSL-KDD dataset modeling
- ■ **Security Domain:** Network protocols, intrusion detection, threat categorization (DoS, Probe, R2L, U2R)
- ■ **Design:** Responsive layout, dark mode theming, CSS animations, semantic color tokens

Key Achievements

- ✓ Architected a modular, component-driven codebase with reusable UI primitives and clean separation of concerns.
- ✓ Simulated a real-world SOC environment with streaming data, making the project demo-ready for interviews and portfolios.

- ✓ Implemented a complete ML evaluation dashboard (accuracy, precision, recall, F1, confusion matrix) without backend dependencies.
- ✓ Built a production-quality dark UI with custom animations, gradients, and a cohesive cybersecurity visual identity.

Tools: React · TypeScript · Tailwind CSS · Recharts · Framer Motion · Vite · Radix UI · shadcn/ui